Young-mok Jung, Ph.D

Machine Learning Research Engineer

bit.ly/youngmokjung
San Diego, CA

WORK EXPERIENCE

OCT 2025 – PRESENT **Apple Inc.** **San Diego, CA**

Machine Learning Engineer

- **RL-post training, Evaluation, and Agentic System:** Developing and evaluating models for agentic systems (LLM-Siri) to handle complex, multi-step user queries. Also implemented harness-agnostic memory evaluation framework focusing on personal assistant task domain.

AUG 2023 – SEP 2025 **Inocras Inc. - bioinformatics startup for precision oncology** **San Diego, CA**

AI & Data Team Lead / AI Tech Lead

- **Genomic Foundation Model:** Spearheaded the development of *DNACHUNKER* (arXiv:2601.03019), a learnable tokenization strategy for DNA language models. Led collaborative research to train and evaluate foundation models on proprietary multi-modal datasets for cancer subtyping.
- **MRDVision:** Led design, implementation, and **clinical validation** of an ML-based Minimal Residual Disease (MRD) detection pipeline, used to evaluate therapeutic efficacy in drug development. Achieved **ppm-level sensitivity** in collaboration with Ultima Genomics.
- **HPC & Cloud Optimization:** Re-architected the cancer genome analysis workflow for cloud efficiency by eliminating centralized disk dependencies, enabling vertical/horizontal scalability, integrating hardware acceleration, and optimizing CPU, memory, and storage utilization. Reduced runtime by **50%** and costs by **80%**, driving a **10%+ product margin increase**.
- **CNS Tumor Classification:** Designed an ML pipeline for whole-genome methylation–based classification of CNS tumors, translating biological domain knowledge into computational objectives.

SEP 2018 – FEB 2024 **KAIST** **South Korea**

Graduate Research Assistant (Advisor: Prof. Dongsu Han & Prof. Young Seok Ju)

- **Transfer Learning for Deep Variant Callers:** Developed a domain adaptation framework for DeepVariant and Clair3. Improved SNP/INDEL F1-scores by **6.4%p / 9.4%p** and maintained accuracy with **50% less labeled data** (Preprint ’23).
- **BWA-MEME (Bioinformatics Acceleration):** Designed memory and CPU-cache efficient alignment software using learned indexes. Reduced CPU instructions by **4.6x** and LLC misses by **2.2x**, achieving **3.45x speedup** over Intel BWA-MEM2 (Bioinformatics ’22, cited 270 times).
- **LiveNAS:** Built a live video streaming system using online training for super-resolution DNNs. Reduced bandwidth usage by **45.9%** while maintaining quality (SIGCOMM ’20, cited 172 times).

EDUCATION

SEP 2018 – FEB 2024 **KAIST** **Daejeon, Korea**

Ph.D. in Electrical Engineering (AI/ML Systems for Bioinformatics)

Dissertation: ***Enhancing genome analysis pipeline with AI and ML***

FEB 2014 – AUG 2018 **KAIST** **Daejeon, Korea**

B.S. in Electrical Engineering

SKILLS

GENERATIVE AI & RL **Foundation Models, Agentic system, Transformers, RL, PyTorch**

ML SYSTEMS & HPC **CPU&Memory profiling, SIMD/C++, RDMA/TCP**

Implemented algorithms to optimize custom transport protocols that reduce latency in datacenter or accelerate alignment software in bioinformatics.

COMPUTATIONAL BIO **Alignment, Mutation Calling, Minimal Residual Disease detection, Methylation sequencing**

Designed clinical-grade pipelines for cancer detection and drug efficacy assessment.

PUBLICATIONS

[P1] DNACHUNKER: Learnable Tokenization for DNA Language Models — *(to appear) ICML '26*
*T. Kim, J. Shin, H. Kim, **Youngmok Jung**, et al.*

[A1] Improving minimal residual disease detection: A tumor-informed whole-genome approach — *ASCO '25*
***Youngmok Jung**, S. Ferguson, S. Lee, et al.*

[P2] Enabling whole-genome DNA methylation-based classification of central nervous system tumors — *medRxiv '25*
*J. Lee, Y. Ju, B. Oh, ... **Corresponding: Youngmok Jung***

[C1] TopFull: An Adaptive Top-Down Overload Control for SLO-Oriented Microservices — *SIGCOMM '24*
*J. Park, J. Park, **Youngmok Jung**, H. Lim, H. Yeo, D. Han*

[P3] Generalizing deep variant callers via domain adaptation and semi-supervised learning — *Preprint '23*
***Youngmok Jung**, J. Park, H. Lim, J. Lee, Y. Ju, D. Han*

[J1] BWA-MEME: BWA-MEM emulated with a machine learning approach — *Bioinformatics '22*
***Youngmok Jung**, D. Han*

[C2] Co-optimizing for Flow Completion Time in Radio Access Network — *CoNEXT '22*
*J. Kim, Y. Lee, H. Lim, **Youngmok Jung**, S. Kim, D. Han*

[C3] Engorgio: Neural Video Enhancement at Scale — *SIGCOMM '22*
*H. Yeo, H. Lim, J. Kim, **Youngmok Jung**, J. Ye, D. Han*

[C4] Towards Timeout-less Transport in Commodity Datacenter Networks — *EuroSys '21*
*H. Lim, W. Bai, Y. Zhu, **Youngmok Jung**, D. Han*

[C5] Neural-Enhanced Live Streaming: Improving Live Video Ingest via Online Learning — *SIGCOMM '20*
*J. Kim, **Youngmok Jung (Co-First)**, H. Yeo, J. Ye, D. Han*

[C6] Enabling Neural-enhanced Video Streaming on Commodity Mobile Devices — *MobiCom '20*
*H. Yeo, C. Chong, **Youngmok Jung**, J. Ye, D. Han*

[C7] Neural Adaptive Content-aware Internet Video Delivery — *OSDI '18*
*H. Yeo, **Youngmok Jung**, J. Kim, J. Shin, D. Han*

AWARDS & HONORS

Feb 2023 — **Samsung 29th Humantech Paper Award** (Silver Prize)
Communication & Networks Track

Feb 2022 — **Samsung PhD Sponsorship**
Selected for full research funding and tuition support

Sep 2020 — **1st Place, Kiwoom US Stock Trading Competition**
Achieved 201% ROI (Ranked 1st out of 10,000+ participants)

OPEN SOURCE & PROJECTS

Main Author — **BWA-MEME (ML-Accelerated Aligner)** — *136* *34k+ Pulls*
Optimized C++ genome aligner using ML based index structures. Achieved **3.45x speedup** over Intel BWA-MEM2. Widely used in production pipelines via Bioconda/Docker.

Main Author — **RUN-DVC (Deep Variant Caller)** — *PyTorch*
Semi-supervised domain adaptation framework for genomic variant callers (DeepVariant, Clair3). Enables accurate mutation calling on new sequencing platforms with limited data.

Contributor — **Collaborative Systems Projects** — *C++, NS-3, TensorFlow*
Contributed to high-performance networking and video systems: **NeuroScaler** (Server-side video enhancement), **NEMO** (Mobile super-resolution), and **TLT-RDMA** (Datacenter transport).